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AgriIntelligence: An Integrated Machine Learning-Based Agricultural Analytics and Decision-Support System for Crop Yield Prediction, Market Price Forecasting, and Weather Risk Assessment

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Abstract **—** Agriculture plays a vital role in food security and economic development, but agricultural planning is affected by changing weather conditions, fluctuations in market prices, and uncertainty in crop yield. This paper presents AgriIntelligence, an integrated agricultural analytics and decision-support system designed to support data-driven crop planning. The proposed system combines crop-yield, weather, and market-price datasets through data preprocessing, integration, exploratory data analysis, and feature engineering. Random Forest machine-learning models are used for crop-yield and market-price prediction. In addition, the system analyzes weather-related factors, including rainfall, temperature, and humidity, to assess potential agricultural risks. Based on the available yield, price, and weather information, a crop-recommendation module provides insights to support crop selection. An interactive dashboard developed using Streamlit brings together data visualizations, model predictions, weather-risk assessments, and crop recommendations in a single interface. The system aims to simplify access to agricultural insights and demonstrate how machine learning can support agricultural decision-making. Its current limitations include dataset coverage and dependence on the quality of available input data. Future enhancements may incorporate real-time weather forecasts, soil characteristics, satellite data, and comparisons with advanced machine-learning algorithms.

I. INTRODUCTION

Agriculture plays a vital role in food security and economic development worldwide. However, agricultural decision-making is increasingly affected by uncertain crop yields, fluctuating market prices, and changing weather conditions. Accurate crop-yield prediction, market-price forecasting, and early identification of weather-related risks are important for helping farmers make informed decisions. Traditional agricultural planning often relies on historical experience and separate sources of information, which can make it difficult to evaluate multiple factors together. These challenges highlight the need for integrated, data-driven, and accessible agricultural decision-support systems.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly algorithms such as Random Forest, have created opportunities to analyze agricultural data and support crop planning. Machine-learning models can identify patterns in historical datasets and generate predictions that assist in understanding crop yields and market-price trends. In addition, weather data, including rainfall, temperature, and humidity, can be analyzed to identify potentially unfavorable agricultural conditions. Integrating these capabilities into a unified platform can improve the accessibility of agricultural insights and support more informed decision-making.

This study presents AgriIntelligence, an integrated agricultural analytics and decision-support system. The key objectives are: (1) to preprocess, integrate, and analyze crop-yield, weather, and market-price datasets; (2) to develop Random Forest-based models for crop-yield prediction and market-price forecasting; (3) to assess weather-related agricultural risks using available environmental factors; and (4) to provide crop recommendations and present analytical insights through an interactive Streamlit dashboard. The study also considers the limitations of the current system and discusses future improvements, including the use of real-world agricultural data, soil information, weather forecasting, and advanced machine-learning techniques.

II. LITERATURE SURVEY

Related Work

This work presents the application of machine learning techniques to agricultural analysis and crop-yield prediction. Machine-learning models can be trained using historical agricultural datasets containing crop-related and environmental information to estimate crop yields. Such predictions can help farmers and agricultural planners understand expected production and make informed decisions regarding crop selection and resource allocation. The integration of predictive analytics into agricultural systems can improve the accessibility of data-driven insights.

This approach also considers the integration of multiple agricultural data sources, including crop-yield records, weather information, and market prices. Combining these datasets provides a broader view of the factors affecting agricultural decisions. An interactive dashboard can present this information through visualizations, predictions, and analytical summaries, helping users interpret agricultural trends in a centralized interface.

Machine-learning methods can also be applied to market-price forecasting by analyzing historical price records and relevant agricultural features. Understanding market-price trends can support crop-planning decisions and help users compare potential market opportunities. However, prediction quality depends on the availability, consistency, and representativeness of the underlying data.

Weather analysis is another important component of agricultural decision support. Factors such as rainfall,

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temperature, and humidity can influence crop growth and agricultural productivity. By examining these variables, an agricultural analytics system can identify potentially unfavorable conditions and communicate possible weather-related risks. Incorporating weather analysis alongside yield and market-price predictions provides a more integrated perspective on crop planning.

The AgriIntelligence project combines these concepts in a single agricultural analytics platform. Its main components include data preprocessing, data integration, exploratory data analysis, feature engineering, crop-yield prediction, market-price prediction, weather-risk assessment, and crop recommendation. The system presents the resulting insights through an interactive Streamlit dashboard. Further development could incorporate real-time agricultural information, soil characteristics, satellite observations, and additional predictive techniques.

Methodology

Feature Engineering

Feature engineering is an important stage in preparing agricultural data for machine-learning models. The purpose of this process is to identify and prepare relevant variables from crop-yield, market-price, and weather datasets for analysis and prediction. Depending on the available data, these variables may include crop type, historical yield, market price, rainfall, temperature, and humidity.

Before feature engineering, the datasets undergo preprocessing to address issues such as missing values, duplicate records, inconsistent data types, and invalid entries. Data from different sources can then be integrated using appropriate common fields, where available. Exploratory data analysis is used to examine distributions, relationships, and trends in the data. These steps help prepare a consistent dataset for subsequent modeling.

Machine-Learning Model Development

Following data preparation and feature engineering, the selected features are used to train Random Forest models for crop-yield prediction and market-price prediction. Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees. It is suitable as an initial model for structured agricultural datasets because it can represent nonlinear relationships between input features and target variables.

The trained models can generate predictions from the input features supplied to them. Their performance should be assessed using appropriate evaluation metrics, such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) for regression tasks. The metrics should be calculated using held-out test data and reported with the actual measured values. No specific performance results are stated here because verified experimental results have not been provided.

Weather-Risk Assessment and Crop Recommendation

The weather-risk component analyzes available environmental variables, including rainfall, temperature, and humidity, to identify conditions that may be unfavorable for agricultural activities. In the current project, this component should be described as a rule-based or threshold-based assessment if it uses predefined conditions rather than a separately trained predictive model.

The crop-recommendation component combines relevant information from yield predictions, market prices, and weather-risk assessments to provide crop-selection insights. This integrated approach aims to help users compare agricultural options using several factors instead of relying on a single variable. The recommendations remain dependent on the quality and coverage of the available data and the rules or scoring method used by the system.

Model Evaluation

The crop-yield and market-price prediction models should be evaluated using appropriate regression metrics. MAE measures the average absolute difference between predicted and actual values, while RMSE gives greater weight to larger prediction errors. R^2 indicates how much of the variation in the target variable is explained by the model relative to a baseline.

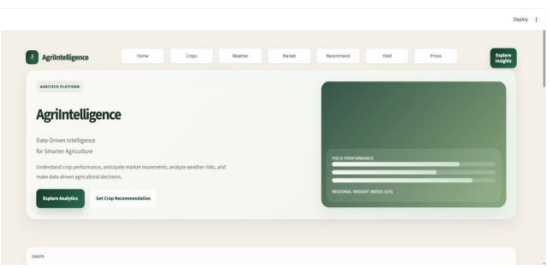
These metrics provide complementary information about prediction performance. Evaluation should use data that were not used to train the models. The resulting scores should be reported only after the models have been tested, so that the reported performance accurately represents the experiments conducted in the project.

Applications and Benefits

By bringing crop-yield prediction, market-price analysis, weather-risk assessment, and crop recommendation together, AgriIntelligence provides a centralized platform for agricultural decision support. The Streamlit dashboard allows users to explore available data, view analytical visualizations, review model predictions, and examine crop recommendations through an interactive interface.

Integrating these components can make agricultural information easier to access and interpret. The platform demonstrates how data analytics and machine learning can support agricultural planning by presenting multiple relevant factors in one place. It may also provide a useful foundation for future systems intended to assist farmers, researchers, and agricultural planners.

However, the current system should be considered a prototype rather than a fully validated agricultural advisory service, particularly if some datasets are synthetic or limited in geographic and historical coverage. Its predictions and recommendations should not be treated as guaranteed outcomes under real-world farming conditions.



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III. DATA RESOURCES

Dataset Collection and Preprocessing

Reliable and diverse data are crucial for developing an effective machine-learning-based agricultural decision-support system. Agricultural productivity and crop-planning decisions are influenced by several factors, including crop yield, weather conditions, and market-price fluctuations. This research focuses on developing AgriIntelligence, an integrated agricultural analytics system that uses available datasets to support crop-yield prediction, market-price forecasting, weather-risk assessment, and crop recommendation.

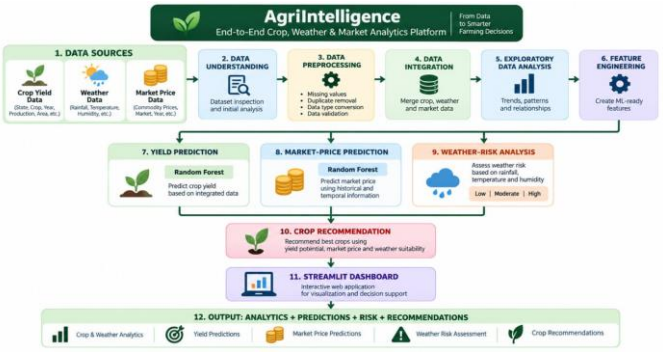
The datasets used in this project contain agricultural information related to crop yield, market prices, and weather conditions. These data provide the foundation for analyzing historical trends and developing predictive models. The integration of these different data sources enables the system to examine agricultural conditions from multiple perspectives. The reliability and representativeness of the datasets are important considerations when interpreting the predictions and recommendations generated by the system.

To improve data quality and prepare the information for analysis, the raw datasets undergo several preprocessing steps. These include handling missing values, identifying and removing duplicate records, converting data types into appropriate formats, and checking data consistency. Where appropriate, the datasets are integrated using common fields to create a unified dataset for further analysis. Exploratory data analysis and feature engineering are then performed to identify relevant variables and prepare the data for machine-learning models.

IV. PROPOSED FRAMEWORK: RETINA SCOPE AI

AgriIntelligence is an integrated agricultural analytics and decision-support system designed to support crop planning through data analysis and machine learning. The system combines crop-yield, weather, and market-price datasets to generate useful agricultural insights. It applies data preprocessing and feature engineering techniques to prepare the datasets for analysis and uses Random Forest models for crop-yield prediction and market-price forecasting. In addition, the framework analyzes environmental factors, including rainfall, temperature, and humidity, to identify potentially unfavorable agricultural conditions.

The system integrates these analytical components into an interactive dashboard developed using Streamlit. Through the dashboard, users can explore agricultural data, examine predictions, review weather-risk assessments, and obtain crop recommendations based on the available information. The framework provides a unified approach to agricultural analysis and demonstrates how machine learning can support data-driven decision-making. The development methodology consists of several key stages.



A. Data Collection and Preprocessing

The AgriIntelligence system uses datasets containing information related to crop yield, weather conditions, and market prices. These datasets provide the foundation for understanding agricultural trends and developing predictive models. Before analysis, the data undergo preprocessing to improve consistency and suitability for machine learning.

The preprocessing stage includes handling missing values, identifying duplicate records, converting data types, and checking data consistency. These operations help reduce data-quality issues that could affect subsequent analysis. Where appropriate, datasets are integrated using common fields to create a unified source for agricultural analytics.

The prepared data are then used for exploratory data analysis and feature engineering. These processes help identify relevant variables, examine relationships among agricultural factors, and prepare suitable input features for model development.

B. Machine-Learning-Based Prediction

The proposed framework uses Random Forest algorithms to develop crop-yield and market-price prediction models. Random Forest is an ensemble learning technique that combines multiple decision trees to produce predictions. It is used in this project as an initial machine-learning approach for analyzing structured agricultural data.

The crop-yield prediction component estimates yield using the available agricultural features. The market-price prediction component analyzes relevant historical data to estimate market prices. These predictions provide information that can assist users in examining crop-production and market trends.

The performance of the models can be evaluated using regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The actual evaluation results should be reported after testing the trained models on appropriate held-out data.

C. Weather-Risk Assessment

Weather conditions can significantly influence agricultural productivity and crop-growing suitability. The AgriIntelligence framework analyzes available environmental variables, including rainfall, temperature, and humidity, to identify potentially unfavorable conditions.

The weather-risk component provides an additional source of information for agricultural planning. Its assessment is combined with the outputs of the prediction components to give users a broader understanding of the factors that may affect crop selection.

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Where risk levels are determined using predefined thresholds or rules, the component functions as a rule-based assessment rather than a separately trained weather-prediction model. Its usefulness therefore depends on the suitability of the selected rules and the quality of the available weather data.

D. Crop Recommendation

The crop-recommendation component combines information from crop-yield predictions, market prices, and weather-risk assessments to provide crop-selection insights. By considering several factors together, the system aims to help users compare agricultural options more systematically.

The recommendations depend on the scoring rules, input features, and data available within the system. They should be interpreted as decision-support suggestions rather than guaranteed predictions of crop profitability or suitability.

E. Interactive Dashboard and Data Visualization

The AgriIntelligence dashboard is developed using Streamlit to provide an accessible interface for interacting with the system. It brings together agricultural data analysis, model predictions, weather-risk information, and crop recommendations.

The dashboard enables users to explore the available information without directly interacting with the underlying machine-learning code. Visualizations help present historical trends, while prediction and recommendation components make the analytical outputs easier to review.

The integrated dashboard serves as the user-facing component of the framework and demonstrates how agricultural data analytics can be organized into a single decision-support platform.

V.RESULT AND DISCUSSION

Fig.3 Basic UI when user enter the web interface Figure 3 depicts the original user interface of the web application for diabetic retinopathy detection. To the left, a loading screen animation or logo is rendered as a visual indicator while the application loads. The AgriIntelligence project demonstrates the integration of agricultural data preprocessing, exploratory data analysis, machine-learning-based prediction, weather-risk assessment, and crop recommendation within a Streamlit dashboard. The framework provides a common interface for examining agricultural information and reviewing the outputs of its analytical components.

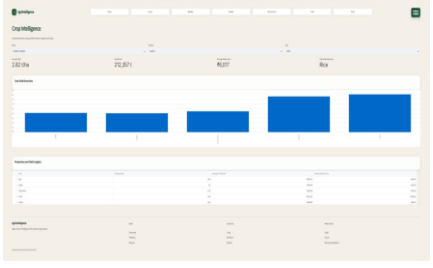


Figure 3. AgriIntelligence Dashboard Interface

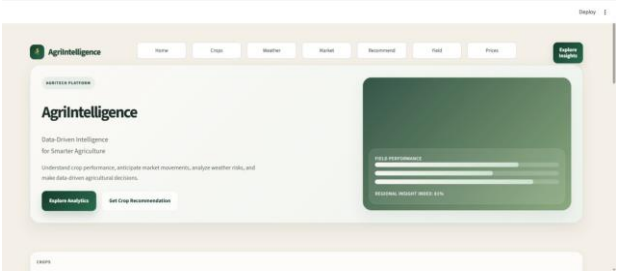


Figure 3 should present the main interface of the AgriIntelligence application. The dashboard serves as the entry point through which users can access the available agricultural analytics and decision-support features. Its layout organizes the system's functionality and allows users to interact with the available data and outputs.

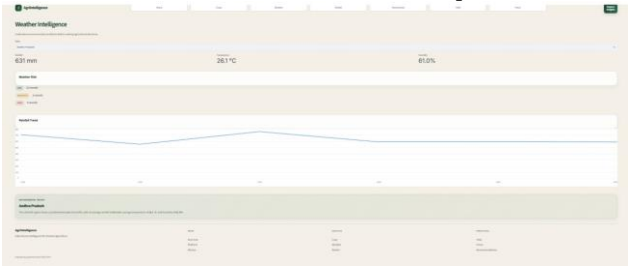


Figure 4. Agricultural Data Input and Analysis

Figure 4 should illustrate the agricultural datasets used by the system or the data-analysis interface. The underlying workflow includes preprocessing and validation steps to prepare the crop-yield, weather, and market-price data for analysis. These steps help establish a consistent data foundation for the prediction and recommendation components.

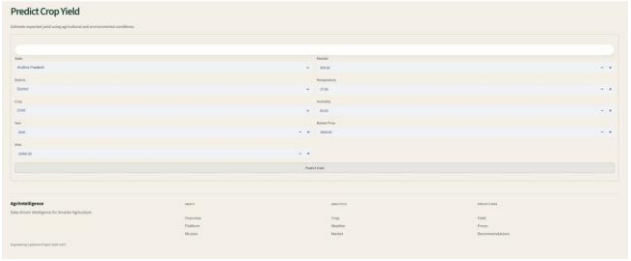


Figure 5. Crop-Yield and Market-Price Prediction

Figure 5 should show the prediction interface or an actual output from the crop-yield and market-price models. The Random Forest models use the available input features to generate predictions. Their reliability should be discussed using the evaluation metrics obtained from testing, rather than assumed from the presence of a prediction interface alone.

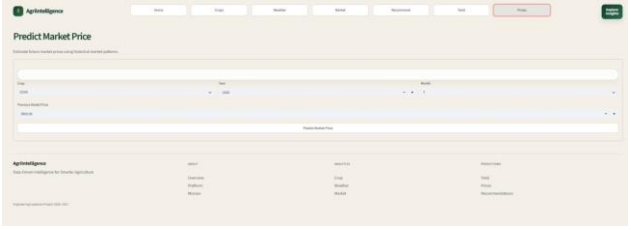


Figure 6. Weather-Risk Assessment and Crop Recommendation

Figure 6 should present the weather-risk assessment and crop-recommendation components. The system considers available environmental conditions alongside yield and market-price information to provide agricultural insights. These recommendations are intended to support comparisons between available crop options and should not be interpreted as guaranteed outcomes.

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Overall, AgriIntelligence demonstrates a unified approach to agricultural analytics by combining data preparation, predictive modeling, environmental assessment, and crop recommendation in one application. The integration of these components provides a foundation for exploring how data-driven methods can assist agricultural decision-making.

However, the practical reliability of the system depends on the quality and representativeness of the datasets, the predictive performance of the trained models, and the suitability of the recommendation rules. If synthetic or demonstration datasets are used, the results should be interpreted as a prototype demonstration and should not be presented as proof of real-world agricultural performance. Further evaluation using representative real-world data is necessary before the system can be relied upon for operational farming decisions.

VI. CONCLUSION AND FUTURE SCOPE

This paper presents AgriIntelligence, an integrated machine-learning-based agricultural analytics and decision-support system that combines crop-yield prediction, market-price forecasting, weather-risk assessment, and crop recommendation. By integrating crop-yield, weather, and market-price datasets, the proposed system provides a unified platform for analyzing agricultural information and supporting data-driven crop-planning decisions. Random Forest models are used for crop-yield and market-price prediction, while environmental factors such as rainfall, temperature, and humidity are analyzed to identify potentially unfavorable agricultural conditions. The system further combines these insights to generate crop recommendations and presents the results through an interactive Streamlit dashboard.

The integration of multiple agricultural analytics components into a single platform makes relevant information easier to access, visualize, and interpret. The dashboard enables users to explore available data, review predictions, assess weather-related risks, and examine crop recommendations within one application. This approach demonstrates the potential of combining machine learning and data analytics to support agricultural planning. However, the reliability and practical usefulness of the system depend on dataset quality, model performance, and the suitability of the recommendation methodology. If synthetic or demonstration datasets are used, further validation with representative real-world agricultural data is necessary before applying the system to actual farming decisions.

Future development should focus on improving prediction reliability, expanding data coverage, and increasing the transparency of the system's recommendations. Incorporating real-time weather forecasts, updated market prices, soil properties, soil moisture, and satellite-derived information could provide a more comprehensive understanding of agricultural conditions. Comparing Random Forest with other machine-learning algorithms, such as Gradient Boosting and XGBoost, could help identify models that perform better on the available datasets. In addition, explainable AI techniques could help users understand which input features influence the model's predictions and recommendations.

Further enhancements may include crop-profitability estimation, location-specific agricultural analysis, and automated alerts for significant changes in weather conditions or market prices. Developing a mobile-friendly interface could also improve accessibility for users in different agricultural settings. Finally, testing the system with larger and more diverse real-world datasets, evaluating its performance across different regions and seasons, and gathering feedback from intended users would help assess its practical value. These improvements could support the evolution of AgriIntelligence into a more reliable, transparent, and comprehensive agricultural decision-support platform.

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